Toy Piano on Cumulonimbus 0.1.2 – Quick Guide

The Toy Piano through a Clouds-style granular cloud and a stereo hall reverb, for the Electro-Smith Seed3 Desktop Development Kit · 12 November 2026 · `main` @ `a97569f` (images built at `9075282`)

> **Status:** host tests, sanitizers and all ARM builds passed. Sound, controls, CPU load and the **direction of SW18** on the real kit are **NOT MEASURED**. Start at a low listening level.

1. Firmware

Image	Use	SHA-256 (start)
ToyPianoCumulonimbus48-v0.1.2.bin	**P48**: the playable instrument, grains at 48 kHz	`5cd63420…`
ToyPianoCumulonimbus32-v0.1.2.bin	**P32**: the same with the grains at 32 kHz (lighter CPU load)	`7d9c7ba9…`

Store: `firmware/toy-piano-cumulonimbus[-32]/0.1.2+9075282/`. Test package with the run sheet, profiling and standalone images and the v0.2 rollback: `build/cumulonimbus-stage2/`. Check the SHA-256 first.

- **Enter the bootloader:** hold **S9 + S16** (and no other key) for **5 s**, or click **Firmware update …** in the editor. All 16 LEDs light up; the bootloader waits without a time limit.
- **Flash:** Daisy Web Programmer → Custom → the `.bin`, or `dfu-util -a 0 -s 0x90040000:leave -D <image>.bin -d ,0483:df11`.

2. Connections and signal flow

Connector	Use
Audio OUT L/R, headphones	Stereo output, 48 kHz; RV1 is the analogue headphone volume
USB J7	USB-MIDI (notes in and out, CCs), the editor, firmware update
MIDI DIN in	Notes and CCs, handled like USB (no SysEx)
CV IN 1/2, Gate IN 1/2	Sources and sync inputs of the Mod on Daisy matrix

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Toy Piano (mono) --> grain cloud (2 s recording, stereo grains) --> hall reverb (stereo) --> OUT
                     Grain mix: dry/grains                          Hall mix: dry/reverb
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With Grain mix and Hall mix at 0, or in FX Bypass, the regular Toy Piano sound comes out unchanged.

3. Playing

- **S1–S8** play C4–C5, **S9–S16** C5–C6 (white keys). Each press starts a one-shot attack/decay envelope; releasing does not stop it. LEDs D1–D16 show the held keys.
- The keys send their notes over USB-MIDI on channel 1. Incoming MIDI notes (any channel, with velocity) play 16 more voices.

4. MIDI

Message	Function
Notes, any channel	Play piano voices (note-offs are ignored)
CC 70–77, ch. 1–15	FX page VR1–VR8 (position, length, density, pitch, grain mix, spread, hall decay, hall mix)
CC 86–90, ch. 1–15	Piano attack, decay, transpose, harmonic, level
CC 1, aftertouch	Sources of the modulation matrix

5. Controls

SW17 only selects which page the eight pots edit. Both the piano and the effects keep running with their settings.

Pot	SW17 off: Piano page	SW17 on: FX page
VR1	Attack, 2–300 ms	Grain position: left = newest audio, right = 2 s back
VR2	Decay, 50–3000 ms	Grain length, 32–512 ms
VR3	Transpose ±24 semitones (centre detent)	Density: left regular, centre none, right random
VR4	Second harmonic (brightness)	Grain pitch ±24 semitones (centre detent)
VR5	Piano level	Grain mix
VR6	Modulation LFO 1 rate	Stereo spread of the grains
VR7	Modulation LFO 2 rate	Hall decay
VR8	Modulation Random rate	Hall mix

Switch	Function
SW18 up	**Freeze**: the recording stops and the grains keep playing the held sound; the reverb keeps running. Releasing cross-fades back to live input.
SW18 centre	**Run**: normal operation
SW18 down	**FX Bypass**: new sound no longer enters the effects, the tail rings out for about 2 s, then the output is the exact dry piano

> Which way is “up” on your kit is checked in the first board run; please note it.

- **Pickup:** after a page change, a MIDI CC or an editor change, a pot does nothing until you turn it to the stored value (or across it). Values never jump.
- **Power-up:** with SW17 off, the Piano page starts at the pot positions. The FX page always starts at gentle defaults (little grain and reverb mix) and picks up as described.

6. Editor and modulation

- Use the Mod on Daisy editor (Daisy Control Panel / VST3 **0.5.1** or the browser editor **0.4.1**); the instrument reports as `toy-piano-cumulonimbus`. Close other programs that hold the MIDI port.
- Matrix targets: Pitch, Level, Harmonic, Attack, Decay, **Grain position**, **Grain mix**, **Hall mix**. VR6–VR8 (Piano page) set the patch's LFO and Random rates; an idle editor shows the change within about 2 s.

7. If something seems wrong

Observation	Check
A pot does nothing	Pickup: turn it to the current value. Check the page (SW17).
No effect audible	SW18 in Bypass? Grain mix (FX VR5) and Hall mix (FX VR8) turned up? Density (VR3) at centre = no grains.
The sound keeps repeating	SW18 is in Freeze.
Crackles or drop-outs	Note the settings; read the load with `python mod_probe.py stats` (in the test package tools); try P32.
Back to a known state	Flash Toy Piano 0.5 or the released v0.2 from the test package.

Sources: instruments/toy-piano/docs/CUMULONIMBUS.md, README.md, effects/cumulonimbus · Repository sscheidl/Daisy-Seed, PR #108 · Hardware results: record with the date as MEASURED.